

Berfin Celik

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Software engineer working across backend systems and machine learning: 2+ years of production .NET Core experience and an MSc in Computer Science & Artificial Intelligence (Distinction). Builds and ships REST APIs, microservices, NLP pipelines, LLM fine-tuning, and RAG systems.

EDUCATION

University of Bradford - MSc Computer Science & Artificial Intelligence | **Distinction** 2024 - 2025

Dissertation: Designed and evaluated an end-to-end NLP pipeline for multidialectal language understanding in low-resource languages (Turkish, Kurmanji, Zazaki), combining ASR, topic modelling, and knowledge graph construction. Fine-tuned Whisper and Wav2Vec2 on a custom corpus; implemented BERTopic, KeyBERT/YAKE, and NetworkX for semantic knowledge graphs. Evaluated with WER, CER, BLEU, and topic coherence, with human-in-the-loop validation by native speakers. ([PDF](#)) ([Code](#))

Bahcesehir University - BSc Computer Engineering 2016 - 2021

EXPERIENCE

Junior .NET Developer - Path Product & Software House Istanbul, Turkey Dec 2023 - May 2024

- Developed a large-scale school management platform end to end for post-pandemic hybrid (online and in-person) teaching, covering course scheduling, learning materials, assessments and grading, and automated question generation.
- Optimised C# / .NET Core backend services through performance tuning and efficient algorithms, and integrated Redis distributed caching to reduce database load and improve response times.
- Implemented the platform's multi-layered microservices architecture based on Onion Architecture, using RabbitMQ for messaging between services, and achieved a modular, testable codebase for quicker feature updates.
- Containerised applications with Docker for consistent environments and reduced downtime; collaborated in cross-functional Agile/Scrum teams, documenting and testing APIs with Swagger.
- Enhanced JavaScript functionality for company web pages, improving client-side responsiveness.

Software Engineer - Argevi Software & Automation Istanbul, Turkey Mar 2022 - Jun 2023

- Built the backend of a real-time industrial automation system for warehouse-scale conveyor logistics using .NET Core and .NET Framework.
- Processed live PLC barcode signals and returned categorisation and routing decisions directing items to the correct chute.
- Developed live monitoring and logging for the automation line, with real-time screens for item flow, vehicle and gate status, plus Elasticsearch, Logstash, and Kibana dashboards.
- Led a 6-month R&D effort that successfully adapted the OPC industrial communication protocol into the company's stack; evaluated Docker adoption and recommended against it given project scale.
- Designed efficient database schemas and optimised queries, improving data processing speed on high-traffic workloads; contributed to a project management tool (ASP.NET Core + React).

IT Intern - Turkuvaz Media Group Istanbul, Turkey Dec 2020 - Mar 2021

- Performed user testing on updated application components, monitored database performance, and assisted in optimising MySQL queries for stable releases.

Student Assistant - Bahcesehir University Engineering Faculty Istanbul, Turkey Nov 2019 - Mar 2020

- Assisted students with coding assignments in C, C++, and Java, clarifying complex programming concepts.

PROJECTS

University Chatbot: LLM Fine-Tuning & RAG - LLaMA 2, DeepSeek, QLoRA, FAISS, Hugging Face ([GitHub](#))

- Scraped SharePoint content into 600+ QA pairs; fine-tuned LLaMA 2 and DeepSeek with QLoRA and implemented FAISS-based RAG retrieval, reducing hallucinations.
- Benchmarked four model variants on retrieval and generation quality; deployed via Hugging Face Spaces.

AI-Powered Deepfake Voice Detection - CNN, Whisper, LoRA, Quantisation, FastAPI, Docker ([GitHub](#))

- Engineered a hybrid CNN + Whisper pipeline for synthetic audio detection using MFCC features and transformer embeddings; applied LoRA fine-tuning and quantisation for efficient multilingual inference.
- Trained and evaluated on 220k+ real and synthetic audio samples; deployed as a FastAPI service containerised with Docker.

Dialectal Word Mining & Knowledge Graph - Whisper, Wav2Vec2, NetworkX, Python ([GitHub](#))

- Mined dialect vocabulary across Turkish, Kurmanji, and Zazaki via ASR; built co-occurrence knowledge graphs with NetworkX.

PROFESSIONAL DEVELOPMENT

ML Specialisation - Coursera (IBM) 2025 - Present

Completed 'Generative AI Language Modeling with Transformers' and 'Introduction to Neural Networks and PyTorch'.

SKILLS

Programming: Python, C#, SQL, Java, C, C++, JavaScript

Backend & APIs: .NET Core / ASP.NET, FastAPI, REST APIs, Microservices (Onion Architecture), Swagger, RabbitMQ, OPC / TCP-IP

ML & Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, LLM Fine-tuning (LoRA, QLoRA), NLP (ASR, RAG), Whisper, Wav2Vec2, Model Quantisation, FAISS, BERTopic, NetworkX, Gradio

DevOps & Monitoring: Docker, CI/CD (GitHub Actions), GCP, Elasticsearch, Logstash, Kibana

Databases & Caching: PostgreSQL, MySQL, SQLite, Redis

Languages: English (Proficient), Turkish (Native)